

WHAT IS LIFE?

Life can be defined as a combination of seven main actions—known as the characteristics of life—that set living things apart from nonliving things. However big or small, every organism must process food, release energy, and excrete its waste. All will also, to some degree, gather information from their surroundings, move, grow, and reproduce.

Life on a leaf

The characteristics of life can all be seen in action on a thumbnail-sized patch of leaf. Tiny insects, called aphids, suck on the leaf's sap and give birth to the next generation, while leaf cells beneath the aphids' feet generate the sap's sugar.

Sensitivity

Sense organs detect changes in an organism's surroundings, such as differences in light or temperature. Each kind of change—called a stimulus—is picked up by them. With this information, the body can coordinate a suitable response.



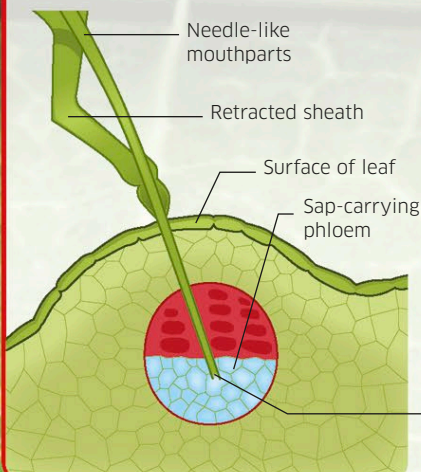
Segments near the end of the aphid's antenna contain sense organs.

Antenna

Aphid antennae carry different kinds of sensors, including some that detect odors indicating a leaf is edible.

Nutrition

Food is either consumed or made. Animals, fungi, and many single-celled organisms take food into their body from their surroundings. Plants and algae make food inside their cells, by using light energy from the sun to convert carbon dioxide and water into sugars and other nutrients.



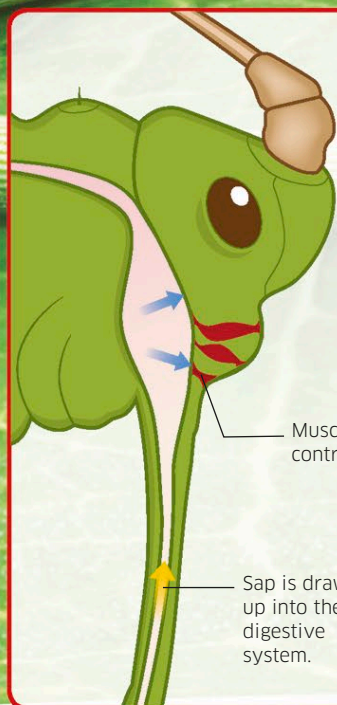
Proboscis

Like most other animals, aphids pass food through a digestive system, from which nutrients move into the body's cells. Aphids can only drink liquid sap. They use a sharp proboscis that works like a needle to puncture a leaf vein to get sap.

Pressure in the leaf's vein forces sap up through the proboscis of the aphid.

Movement

Plants are rooted in the ground, but can still move their parts in response to their surroundings—for instance, to move toward a light source. Animals can move their body parts much faster by using muscles, which can even carry their entire body from place to place.



Head muscles

Muscles are found all over an aphid's body. As the aphid eats, muscles in its head contract (shorten) to pull and widen its feeding tube. This allows it to consume the sap more effectively.

